

Binayak Kunwar

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Department of Agricultural Economics, Purdue University | West Lafayette, IN

Education

Purdue University <i>Ph.D. in Agricultural Economics</i>	May 2027 (expected) West Lafayette, IN
Purdue University <i>M.S. in Agricultural Economics</i>	August 2024 West Lafayette, IN
Agriculture and Forestry University <i>Bachelor of Science in Agriculture</i>	December 2021 Rampur, Nepal

Dissertation

Committee: Dr. Chad Fiechter (Chair), Dr. Guy Tchuente, Dr. Michael Langemeier, and Dr. Tim Cason

Dissertation Chapters

- The Economic Development Premium in Farmland Markets
- When Solar Projects Fail: Belief Reversal and Farmland Prices
- Market Structure, Information, and Market Power in Farmland Transactions

Academic Profile

Ph.D. candidate in the Department of Agricultural Economics at Purdue University specializing in farmland and land-use economics, agricultural finance, renewable energy development, and applied econometrics. Research emphasizes spatial markets, causal identification, and policy-relevant land-market outcomes.

Research Interests: Agricultural Finance, Farmland and Land-Use Economics, Production Economics, Technology Adoption, Risk Management, and Applied Econometrics

Academic and Research Experience

Department of Agricultural Economics, Purdue University <i>Research Assistant</i>	Fall 2024 – Present West Lafayette, IN
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- Conduct research on U.S. corn producers' use of over-the-counter risk management products in collaboration with Dr. Chad Fiechter, Dr. Mindy Mallory, Eugene Faller, and Marex Solutions; presented findings at the 2025 AAEE and WAEA Joint Annual Meeting.
- Estimate cross-return elasticities of farmland conversion to renewable energy with Dr. Chad Fiechter and Dr. Guy Tchuente; presented findings at the 2025 NC1177 Annual Meeting.
- Analyze the effects of Indiana Economic Development Corporation land acquisitions on farmland prices and spatial spillovers.
- Design a laboratory experiment to study information, market structure, behavioral responses, and market power in farmland transactions.
- Presented research on ESG practices and financial performance in agribusinesses at the 2024 NC1177 Annual Meeting.

Department of Agricultural Economics, Purdue University <i>Research Assistant</i>	Fall 2022 – Summer 2024 West Lafayette, IN
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- Conducted data collection, cleaning, spatial analysis, and visualization for research on farmland prices and solar energy development under the supervision of Dr. Todd Kuethe.

- Supported data cleaning, empirical analysis, and graphics development for applied agricultural economics research projects led by Dr. Chad Fiechter.
- Presented findings from “Impacts of Commercial and Utility-Scale Solar Energy on Farmland Prices” at the NC1177 Annual Meeting.

**Prime Minister Agriculture Modernization Project (PMAMP) February 2021 – June 2021
Kaski**

Agriculture Intern

Pokhara, Nepal

- Supported agricultural extension and research activities, including producer training, household surveys, profile book preparation, subsector business analysis, package-of-practices preparation, field monitoring, and evaluation.

Teaching Experience

Department of Agricultural Economics, Purdue University

Fall 2026 (planned)

Course Module Development and Instruction

West Lafayette, IN

- AGECE 68400: Applied Quantitative Methods. Developing a time-series analysis module, lectures, and assignments for MS-MBA students.
- AGECE 41100: Farm Management. Developing and teaching a hands-on farmland valuation module.
- AGECE 26100: Strategy and AI in Society, Markets, and Decision Systems. Developing an artificial intelligence module focused on economic, market, and decision-making applications.

Department of Agricultural Economics, Purdue University

October 2024

Instructor

West Lafayette, IN

- AGECE 68400: Applied Quantitative Methods. Delivered a Power BI data visualization session to MS-MBA students.

Department of Agricultural Economics, Purdue University

Fall 2022

Teaching Assistant Support

West Lafayette, IN

- AGECE 20300: Introduction to Microeconomics for Food and Agribusiness. Assisted with exam preparation and grading.

Research Mentorship

- Served as Graduate Senior Research Mentor and co-advised Shwastika Baral’s master’s thesis and Jackson Lusk’s undergraduate research, providing guidance on research design, data analysis, and applied economics research.

Publications and Thesis

- **Kunwar, B.** (2024). *Impact of Commercial and Utility-Scale Solar Energy on Farmland Price*. Master’s thesis, Purdue University Graduate School.
- **Kunwar, B.** (2024). Resource use efficiency of tomato production under plastic house in Pokhara Metropolitan City, Nepal. *Journal of Agricultural Science and Technology*, 26(5).
- Dhakal, N., **Kunwar, B.**, Subedi, K., & Adhikari, D. (2023). Financial feasibility analysis of sweet orange production in Sindhuli, Nepal. *Food & Agribusiness Management*, 4(2), 100–103.
- Bhandari, K. B., **Kunwar, B.**, Kunwar, R. M., Bussmann, R. W., & Paniagua-Zambrana, N. Y. (2021). *Rhamnus purpurea* Edgew. and *Rhamnus virgata* Roxb. Rhamnaceae. In R. M. Kunwar, H. Sher, & R. W. Bussmann (Eds.), *Ethnobotany of the Himalayas*. Ethnobotany of Mountain Regions. Springer, Cham. https://doi.org/10.1007/978-3-030-45597-2_202-1

Revise & Resubmit

- **The Impact of Commercial and Utility-Scale Solar Energy on High-Value Farmland.** Revise and Resubmit at *Agricultural Finance Review*. [Manuscript](#)

Working Papers

- **Identification of Margin-Specific IV Elasticities in Unordered Outcome Models: Evidence from Farmland Conversion to Renewable Energy.** With Dr. Chad Fiechter and Dr. Guy Tchuente. [Manuscript](#)
- **Farmer Corn Marketing and Over-the-Counter Derivatives.** With Dr. Chad Fiechter, Dr. Mindy Mallory, and Eugene Faller. [Manuscript](#)
- **The Economic Development Premium in Farmland Markets.** With Dr. Chad Fiechter, Dr. Guy Tchuente, and Dr. Aaron Shew. [Manuscript](#)
- **When Solar Projects Fail: Causal Evidence on Farmland Prices.** With Dr. Chad Fiechter and Dr. Aaron Shew. [Manuscript](#)
- **Market Structure, Information, and Market Power in Farmland Transactions.** With Dr. Chad Fiechter and Dr. Timothy Cason. [Proposal](#)
- **The Impact of ESG Practices on Financial Performance in Agribusiness.** With Dr. Chad Fiechter. [Manuscript](#)
- **Farm Income Elasticity of Land-Use Change in the United States: Evidence from the Intensive Margin.** With Dr. Chad Fiechter and Dr. Guy Tchuente.
- **Farmland Price Prediction Using Machine Learning Approaches: A Comparative Analysis of Regression and Ensemble Models.** Independent research.
- **Determinants of Booms and Busts in the U.S. Farmland Market.** Independent research.

Extension and Outreach Articles

- **Kunwar, B., Fiechter, C., Tchuente, G., & Shew, A. (2026).** State-led land acquisition and farmland prices: Evidence from Indiana. *Purdue Agricultural Economics Report, Graduate Student Research Issue*. May 2026.
- **Kunwar, B., & Kuethe, T. (2026).** 2026 farmland and cash rent outlook. *Purdue Agricultural Economics Report, Outlook 2026*. March 2026.
- **Kunwar, B., & Fiechter, C. (2025).** Market trend: December 2025 corn futures: 6-month trend overview. *Indiana Corn Research and Updates, Purdue Corn Agronomy*, Issue 5. May 23, 2025.
- **Kunwar, B. (2024).** Firm financial health of publicly traded agribusinesses. *Purdue Agricultural Economics Report*. May 2024.
- **Kunwar, B. (2024).** Research spotlight: The impact of solar energy on Indiana farmland prices. *Purdue Agricultural Economics Report*. August 2024.

Professional Activities

Conferences and Presentations

Kunwar B., Fiechter C., Tchuente G., and Shew A. (2026) State-Led Land Acquisition and Farmland Prices
2026 AAEA Annual Meeting, Kansas City, MO July 28, 2026

Kunwar B., Fiechter C., and Tchuente G. (2026) Monetary Incentives, Landowner Preferences: Estimating Cross-Elasticities in Farmland Conversion to Renewable Energy
2026 AAEA Annual Meeting, Kansas City, MO July 28, 2026

Kunwar B., Fiechter C., Tchuenté G., and Shew A. (2026) Industrial Development and Farmland Market Repricing: How Public Land Purchases Create Direct and Spillover Effects
SAEA 2026 59th Annual Meeting, Louisville, KY February 2, 2026

Kunwar B. (2025) *Farmland Market and Renewable Energy*
Invited Presentation, Purdue Farm Policy Study Group, Purdue University December 9, 2025

Kunwar B., Fiechter C., and Tchuenté G. (2025) Monetary Incentives, Landowner Preferences: Estimating Cross-Elasticities in Farmland Conversion to Renewable Energy
NC1177 Annual Meeting, Federal Reserve Bank of Kansas City – Omaha Branch October 1–2, 2025

Kunwar B., Fiechter C., Mallory M., and Faller E. (2025) An Evaluation of Corn Farmers' Use of Over-the-Counter Risk Management Products
2025 AAEA and WAEA Joint Annual Meeting, Denver, CO July 27–29, 2025

Kunwar B., Fiechter C. (2025) As-Applied Production Maps for On-Farm Trials
CCA Advisory Council Meeting, Purdue University February 7, 2025

Kunwar B., Fiechter C. (2024) Yield Monitor Data and On-Farm Research
IDAAS networking and poster session, IDAAS, Purdue University December 11, 2024

Kunwar B., Fiechter C. (2024) The Impact of ESG Practices on Financial Performance in Agribusiness
NC1177 Annual Meeting, Federal Reserve Bank of St. Louis, MO October 15–16, 2024

Kunwar B., Kuethe T., Fiechter C. and Langemeier M. (2024) Impact of Commercial and Utility-Scale Solar Energy on Farmland Price
Master's Thesis Defense, Department of Agricultural Economics, Purdue University May 20, 2024

Kunwar B. (2024) Understanding the booms and busts in U.S. farmland market
Graduate Student Poster Competition, Purdue University April 5, 2024

Kunwar B., Giri A. and Subedi D. (2024) Participation of Agricultural Producers in Government Programs
Agriculture Policy Scholars Conference 2024, New Orleans, LA January 13–14, 2024

Kunwar B., Fiechter C. and Kuethe T. (2023) Impact of Commercial and Utility-Scale Solar Energy on Farmland Price
NC1177 Annual Meeting, Federal Reserve Bank of Atlanta, GA October 12–13, 2023

Media

- *Binayak Kunwar – Graduate Ag Research Spotlight*. Featured Profile, Purdue University College of Agriculture, Agricultural Research and Graduate Education. March 2026
- *Shining a Light on Solar Energy on Farmland with Binayak Kunwar* (PAEP 115). Purdue Agricultural Economics Podcast, Purdue University. October 2025

Referee Service

- *Land Use Policy*, Reviewer
- *Agricultural Finance Review*, Reviewer

- *Purdue Agricultural Economics Report*, Graduate Student Section, Reviewer
- *AAEA Annual Meeting Abstracts*, Reviewer

Awards and Achievements

- Boehlje Agribusiness Management Fellowship, Agricultural & Applied Economics Association (AAEA) July 2026
- Outstanding Master's Thesis Award, Agricultural & Applied Economics Association (AAEA) July 2025
- Outstanding Master's Thesis Award, Purdue University April 2025
- Butz Supplemental Scholarship, Purdue University Department of Agricultural Economics 2024–2028
- People's Choice Award – Graduate Student Poster Competition, Department of Agricultural Economics, Purdue University April 2024

Leadership Roles

- Agricultural & Applied Economics Association (AAEA), Member 2023 – present
- Southern Agricultural Economics Association (SAEA), Member 2025 – present
- Purdue Agricultural Economics Graduate Student Organization, Intramural Chair 2023 – 2024
- Junior Red Cross Society, President 2014

Online Courses

- Environmental, Social and Governance Course (Bloomberg) May 2024
- Bloomberg Market Concept Course (Bloomberg) April 2024
- Bloomberg Finance Fundamental Course (Bloomberg) March 2024
- Analyzing Company Performance using Ratios (Coursera Project Network) March 2024

Skills and Interests

Languages

- Nepali: **Mother Tongue**
- English: **Fluent**
- Hindi: **Fluent**

Research and Computer Skills

- Empirical Methods: **Causal Inference, Instrumental Variables, Panel Data, Spatial Analysis, Machine Learning**
- Programming and Statistical Software: **R, Stata, Python, MATLAB, SPSS**
- Other Tools: **QGIS, Tableau, Microsoft Office, Bloomberg Market Concepts, L^AT_EX**